Video Transcript

Video 6: Bayes Theorem Exercise (02:19)

So, let's see if we can calculate what percentage of rich people are happy. The Harvard students said 10% of happy people are rich. So, let's draw our diagrams here. Here, we've got everyone, all people. The blue area are happy people and then the red area are rich people and then we have an intersection of rich and happy. Now that intersection can look like this on the left or it could look like this on the right and we don't have enough information to decide. But using Bayes theorem we can get an answer if we know a couple of extra things.

So, the MIT student thought for a moment and then googled the following. Of all the population, what percentage are happy? And google said 40% are happy. So, now we know that of all the people, H is 40%. And now he googled, what percentage of all people are rich? And, the answer was 5%. So, of all the people, the red area is 5%. So, what can we conclude from this? What do we want to know? Well, what we want to know is, given that we're rich, what's the probability that we're also happy? So, the prior is we are rich, and now the question is, are we happy? So, we want in some sense P H given R, if R is rich, what's the probability we're happy, H? So, go away and see if you can calculate now what percentage of rich people are happy. So, l will give you the answer in the next video. Bye for now.

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